

Place Your Bets: Casino Openings and County Unemployment in the United States

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Abstract

This paper examines the relationship between casino openings and county unemployment in the United States using a county-level panel dataset covering 112,569 observations from 1990 to 2024. The analysis is motivated by the ongoing debate over whether casinos promote local economic development by creating more jobs or simply shift economic activity within a region. To estimate this relationship, the study uses a Two-Way Fixed Effects (TWFE) model that controls for county-specific characteristics and national trends over time. The results indicate a positive and statistically significant relationship between casino operation and county unemployment, with an estimated increase of approximately 0.42 percentage points following casino openings. These findings suggest that casino development alone may not reduce unemployment and should not be viewed as a guaranteed strategy to improve local labor market outcomes.

I. Introduction and Background

Casino development has long been debated as a tool for local economic development and regional revitalization, since supporters argue that casinos generate construction activity, create permanent jobs, and increase tourism, making them an attractive policy option for communities seeking economic growth. In many rural or economically distressed counties, casinos are viewed as a means to increase employment opportunities and generate additional tax revenue.

At the same time, economists have questioned whether these benefits translate into sustainable improvements in local labor markets. Critics argue that casinos may simply shift consumer spending away from other local businesses rather than creating new economic activity. If spending is diverted from restaurants, retail stores, or other entertainment venues, employment gains at casinos may be offset by the job losses elsewhere in the local economy.

This paper examines the relationship between seven selected casino openings in the Midwest and county unemployment using a county-level panel dataset covering the United States from 1990 to 2024. A Two-Way Fixed Effects (TWFE) model is used to compare unemployment outcomes before and after casino openings while controlling for county-specific characteristics and nationwide economic conditions. By using panel data and fixed effects, the analysis attempts to isolate the relationship between casino openings and local labor market outcomes.

Literature Review and Previous Research

The economic effects of casino gambling have been widely studied, with researchers reaching mixed conclusions about its impact on local economies. Early studies often relied on economic impact models that assumed casino spending generated new economic activity,

resulting in optimistic estimates of employment growth. These studies, however, frequently overlooked the possibility that casinos simply shifted consumer spending away from other local businesses.

More recent research has adopted stronger empirical methods, including panel data and quasi-experimental designs, to better identify the causal effects of casino openings. Cotti (2008) found that casinos increased employment in the hospitality and entertainment sectors but also documented evidence of displacement in other industries. Similarly, Evans and Topoleski (2002) examined Native American casinos and found that while casinos generated local economic activity, the overall labor market effects varied across regions.

This paper builds on this literature by applying a Two-Way Fixed Effects (TWFE) model to a county-level panel dataset covering the United States from 1990 through 2024. By controlling for both county-specific characteristics and national economic conditions, the analysis provides additional evidence on the relationship between casino openings and local unemployment.

II. Theoretical Issues and Hypotheses

Economic theory provides competing predictions regarding the effect of casino openings on local unemployment. On one hand, casinos create new employment opportunities through construction, gaming operations, hospitality, food services, and administration. As labor demand increases, unemployment may decline if local residents fill these positions. This prediction is referred to as the Job Creation Hypothesis.

On the other hand, casinos may generate limited net employment if they primarily shift consumer spending away from existing businesses. Instead of creating new economic activity,

households may substitute casino spending for spending at restaurants, retail stores, or other entertainment venues. In this case, employment gains at casinos could be offset by job losses in other sectors, resulting in little or no improvement in overall labor market conditions. This prediction is known as the Displacement (Cannibalization) Hypothesis.

Given these competing theories, the empirical analysis estimates the relationship between casino openings and county unemployment using a Two-Way Fixed Effects model. The direction and magnitude of the estimated coefficient provide evidence on which hypothesis is more consistent with the observed data.

III. Data Description

The analysis uses a county-level panel dataset covering the United States from 1990 through 2024, resulting in 112,569 county-year observations. The primary source for the dependent variable is the Bureau of Labor Statistics (BLS) Local Area Unemployment Statistics (LAUS), which provides annual county-level unemployment rates. These data are widely used in labor market research because they combine household survey information with administrative records to produce consistent unemployment estimates across counties.

The casino data were compiled by identifying the opening year of seven selected casino locations in Michigan, Illinois, Indiana, and Ohio. The analysis focuses on the annual unemployment rates of the counties in which these casinos opened. Each casino was matched to its corresponding county, and a treatment indicator was created that equals one for county-years in which a casino was operating and zero otherwise. All remaining counties in the panel serve as the comparison group throughout the sample period.

The final dataset was created by merging the unemployment and casino data using county identifiers and organizing the observations into a county-year panel. This structure allows the analysis to compare unemployment outcomes across counties while following changes over time. The resulting panel provides sufficient variation to estimate the relationship between casino openings and unemployment using a Two-Way Fixed Effects model.

Table 1 presents the descriptive statistics for the variables used in the analysis. The unemployment rate is measured as a percentage, while the casino variable is a binary indicator equal to one if a casino is operating in a county during a given year and zero otherwise.

Table 1: Summary Statistics

Variable	Mean	Min	Max
Unemployment	6.03	0.4	40.6
Casino	0.0012	0	1

Notes: Unemployment is measured as a percentage. Casino is a dummy variable equal to 1 if a casino is operating in a county-year. Sample includes 112,569 observations.

IV. Empirical Model and Estimation Method

To estimate the relationship between casino openings and county unemployment, this study uses a Two-Way Fixed Effects (TWFE) model. The model controls for both county-specific characteristics that remain constant over time and national economic shocks that affect all counties simultaneously. The empirical specification is:

$$Unemployment_{it} = \beta_0 + \beta_1 Casino_{it} + \alpha_i + \delta_t + u_{it}$$

where Unemployment represents the county unemployment rate, Casino is a binary variable equal to one if a casino is operating in a county during a given year, α_i represents county fixed effects, δ_t represents year fixed effects, and u_{it} is the error term.

County fixed effects control for characteristics that do not change over time, such as geography, historical economic conditions, and long-run differences across counties. Year fixed effects account for nationwide events that influence unemployment, including recessions and changes in overall economic conditions. Together, these controls help isolate the relationship between casino openings and unemployment by accounting for both cross-sectional and time-series variation.

Because observations within a county are likely to be correlated over time, standard errors are clustered at the county level. Clustering provides valid statistical inference in the presence of heteroskedasticity and serial correlation within counties and is a standard approach in panel data analysis.

V. Discussion of Results

The results of the Two-Way Fixed Effects (TWFE) model are presented in Table 2. The estimated coefficient on the casino variable is 0.422, with a clustered standard error of 0.141. This estimate is statistically significant at the 1 percent level, indicating a positive relationship between casino operation and county unemployment after controlling for both county and year fixed effects.

Table 2: Effect of Casino Openings on Unemployment

Variable	(1) Unemployment
casino	0.422*** (0.141)
Constant	6.398*** (0.0302)
Observations	112,569
R-squared	0.513

Notes: Robust standard errors clustered at the county level are reported in parentheses. The dependent variable is the unemployment rate. The variable casino equals 1 if a casino is operating in a county-year. County and year fixed effects are included. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The positive coefficient suggests that counties with an operating casino experienced unemployment rates that were, on average, approximately 0.42 percentage points higher than comparable counties without an operating casino. While this finding is statistically significant, it should not be interpreted as definitive evidence that casinos directly increase unemployment. Instead, the results indicate that casino development alone may not be sufficient to reduce unemployment, and that other local economic conditions may continue to play an important role in determining labor market outcomes.

The year fixed effects capture changes in nationwide economic conditions throughout the sample period. As expected, the coefficients for 2009 and 2010 are positive and highly significant, reflecting the increase in unemployment during the Great Recession. These results demonstrate that broader macroeconomic conditions had a substantial impact on county unemployment regardless of whether a county contained a casino.

The model produces a within R-squared of 0.513, indicating that approximately 51.3 percent of the within-county variation in unemployment is explained by the model. Although the model explains a considerable share of the variation in unemployment, local labor markets are influenced by many factors that are not included in this analysis, such as changes in industry composition, population growth, and local economic policy.

Finally, standard errors are clustered at the county level to account for heteroskedasticity and serial correlation within counties over time, providing more reliable statistical inference. Overall, the results suggest a statistically significant positive association between casino operation and county unemployment, indicating that casino development should not automatically be viewed as a guaranteed strategy for reducing local unemployment.

VI. Conclusion and Directions for Further Research

This study examined the relationship between casino openings and county unemployment using a county-level panel dataset covering the United States from 1990 through 2024. By estimating a Two-Way Fixed Effects (TWFE) model, the analysis controlled for both county-specific characteristics and nationwide economic conditions that could influence unemployment over time.

The results suggest a positive and statistically significant relationship between casino operation and county unemployment. While these findings do not necessarily imply that casinos directly increase unemployment, they suggest that casino development alone may not be sufficient to improve local labor market outcomes. Other economic conditions, including industry composition, local business activity, and broader macroeconomic trends, likely continue to play an important role in determining county unemployment.

These findings contribute to the ongoing discussion surrounding casino development as a tool for local economic growth. Although casinos may generate employment opportunities within the gaming and hospitality industries, the overall effect on county unemployment appears to be more complex than often assumed. Policymakers should therefore consider casino development as only one component of a broader economic development strategy rather than a guaranteed solution for reducing unemployment.

Future research could examine whether the effects of casino openings differ across urban and rural counties or vary according to local economic conditions. Additional work could also investigate outcomes beyond unemployment, such as labor force participation, wages, business formation, or population growth. Expanding the analysis to include additional casino locations and alternative empirical approaches would provide a more complete understanding of the long-run economic effects of casino development.

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